



Yash Agrawal
UG (II Year II Semester)
B.Tech. (Chemical Engineering)
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Area of Interest

Backend Engineering, Systems Engineering, Blockchain Development, Database Internals,

Education

Year	Degree/Examination	Institution/Board	CGPA/Percentage
2026	B.Tech. 2nd Year	Indian Institute of Technology, Roorkee	8.281
2024	Intermediate (Class XII)	Sant Gyaneshwar model school	92.00 %
2022	Matriculate (Class X)	Delhi Public School Panipat Refinery	91.00 %

Projects

FluxDB: a postgres inspired index-organized database | Blocsoc IIT Roorkee February 2026 - June 2026

- Designed the durability system which is a PostgreSQL style redo-only recovery, write-ahead log, and vacuum model
- Implemented MVCC; per-tuple xmin/xmax versioning with a in-memory commit-log (CLOG) visibility check
- Created shared Buffer Pool with CLOCK replacement policy and implemented B-Link Tree for concurrency

MatchBench: Distributed load generator | Blocsoc IIT Roorkee May 2026 - June 2026

- MatchBench is a distributed benchmarking platform for exchange matching engines, executing contestant submissions in isolated sandboxes and evaluating latency, throughput, and market-correctness under realistic order flow.
- Built an eBPF-based measurement pipeline that measures kernel-level timestamps with **~100ns** accuracy.
- Developed a horizontally scalable load generator which performs upto **~360k/s** on a single **4 vCPU** node.
- Architected a Kafka-backed event-driven pipeline spanning sandbox orchestration, telemetry aggregation, validation service with Grafana and Prometheus based observability layer

Deimos: Mobile ZK Benchmarking platform | Blocsoc IIT Roorkee September 2025 - May 2026

- Developed an open-source benchmarking suite for measuring the performance of zero-knowledge proving systems on mobile hardware across latency, memory, storage, and energy metrics.
- Integrated Rust-based proving backends through native FFI bindings to execute proofs directly on a Flutter application.
- Built Node.js/Express REST API backed by PostgreSQL and built Next.js dashboard for cross-framework benchmark visualization; deployed with Docker and Nginx reverse proxy

Orbital - A Multi-Dimensional AMM | Blocsoc IIT Roorkee September 2025 - October 2025

- Orbital is a multi-token Automated Market Maker(AMM) inspired by Paradigm research enabling efficient swaps across multiple assets within a unified liquidity pool for stablecoins
- Developed high-precision Q96.48 fixed-point arithmetic and Newton-Raphson solver in Arbitrum Stylus for gas efficiency which achieved 80% gas reduction as compared to solidity based contracts.
- Built Solidity gateway contracts providing a stable external interface to a Rust-powered Stylus execution engine.

EdgeProxy | Blocsoc IIT Roorkee February 2026 - March 2026

- Built a L7 reverse proxy in Rust(tokio+hyper) supporting host/path-based routing, HTTP caching and request timeouts.
- Implemented a high-concurrency cache using moka (W-TinyLFU) with request collapsing to eliminate thundering herd, and a two-tier L1/L2 architecture backed by persistent disk storage via RocksDB
- Designed per-IP rate limiting with DashMap + leaky buckets with configurable load balancing strategies

Awards / Scholarships / Academic Achievements

- AIR **4347** (JEE ADVANCED) | AIR **7373** (JEE MAINS)
- Winner** at Arbitrum Open House offline hackathon in banglore.
- Winner** at Stack2Deep 2025 in freshers track among 50+ teams
- 2nd Place** at Arbitrum Open House Online hackathon among **200+** teams and got invited to the offline round
- Winner** at **ETH Vietnam** in zircuit track among **100** teams
- Polkadot AssetHub** : Selected among the top 20 teams in the online round and secured **2nd place** in the offline finals.

Skills

Computer languages	Rust, C++, Javascript, Python
Software Packages	Tokio, Hyper, Git , Docker, Linux, PostgreSQL, MERN, kafka, Prometheus, Grafana
Languages Known	English, Hindi

Positions of Responsibility & Extra Curriculars

Member | Blocsoc IIT Roorkee March 2025 - Present

- Won **5** hackathons as a team lead with international wins in **ETH Vietnam** , **Arbitrum Open House** and participated in **10+** global hackathons.
- Contributed to club project Deimos which recieved grant from ethereum foundation.
- contributed to open source projects and lead FluxDB with the goal of learning database and postgres internals.
- Regularly conducted hackathons, public lectures, and competitions to foster the technical culture on the campus.